

One analyst and ten agents

What I learned writing equity research on my own, building a ten-agent AI version of the same process, and running both on the same stocks

Over the summer of 2026 I wrote equity research reports on nine companies with help from an AI model, then built a second version of the process in which ten AI agents gather evidence, argue and rule on each call. I ran both on Microsoft and Coca-Cola. They agreed on the rating both times, and the agents caught mistakes the single process missed. A third test on Alphabet in September produced the first disagreement: the agents called a SELL where I had a HOLD. This paper covers what I built, what it cost, what went wrong, and what I'll check when the calls come due in 2027.

Vidwaan Singhania

BCom student, Gustavson School of Business, University of Victoria, graduating 2027

Code: github.com/Vidwaansinghania/Multi-Agent-Equity-Research

Feedback: linkedin.com/in/vidwaansinghania

Current holdings: blossomsocial.com/users/Vidwaan-Singhania__d7oVAMV5EWqGMPon

A student project, written for learning. Nothing in it is investment advice. I own some of the stocks discussed; the disclosure is on the last page.

1 Overview

I'm a business student at the University of Victoria. Between June and August 2026 I wrote equity research reports on nine public companies in my own time, working with Claude, Anthropic's AI model, inside a note-taking app called Obsidian. I call that process Method 1.

In August I built a second version of the same process, which I call Method 2. Instead of one line of reasoning, ten AI agents split the work. Four gather evidence, one values the company, two argue for and against that valuation, a judge makes the call, one checks the call against my own small portfolio, and one writes the report. I then ran both methods on the same companies and compared what came out: Microsoft and Coca-Cola in August, and Alphabet in September.

This paper explains what I built, what the comparison showed, what it cost, where the project stands and what I plan to do next. I'm not a finance professional, and I expect people who do this for a living will find things I got wrong. That is one reason I'm publishing it.

9

companies covered by Method 1, June to August 2026

10

AI agents in every Method 2 run

2 of 3

head-to-head tests where both methods gave the same rating

16 of 30

challenges from the bull and bear the judge accepted in whole or part

\$71–88

cost of one Method 2 run at API prices

0 of 13

open calls scored so far; the first review is June 20, 2027

The short version

- On Microsoft and Coca-Cola both methods gave the same rating, HOLD, with price targets 2.6% and 5.6% apart. On Alphabet they disagreed: Method 2 said SELL at \$268.81 where Method 1 said HOLD at \$340, off the same closing price.
- That agreement proves less than it sounds like. I built both methods, they follow the same rules, and the same AI does the reasoning in each.
- The agents were more useful for catching mistakes than for changing the answer. They found arithmetic errors in their own valuation, a beta figure taken from the wrong company, and a currency cost that made one of my Method 1 buy instructions lose money.
- On Coca-Cola the two methods reached HOLD from opposite sides of one input, beta. Put Method 1's beta into Method 2's model and the rating becomes SELL.
- A Method 2 run cost about \$74 to \$88 at API prices, two to three and a half times a Method 1 report. More than half of that went on agents re-reading their own instructions.
- The Alphabet disagreement comes from two compounding gaps: Method 2 built its own earnings estimate 11.6% below consensus, then applied a multiple two points lower.
- None of the thirteen open calls can be checked against the market until June 2027, so I can't yet say whether either method is any good.

How this paper is organised

Twenty sections in four parts, then references and two appendices. If you only have a few minutes, read section 1, section 11 and section 17.

PART ONE: THE PROJECT

1	Overview	What I did and what I found
2	Goal and aims	Why I started, what I set out to do, and what success looks like
3	Background and terms	About me, my portfolio and the terms used
5	What I built on	The framework I borrowed from and the research standard I copied

4Tools usedEvery tool the project runs on, and the model versions

PART TWO: THE TWO METHODS

6	Method 1: writing reports on my own	How I wrote nine reports and what I learned from them
7	Method 2: ten agents	The agents, the rules they follow and the record they produce
8	How I compared the two methods	The methods side by side, and how the comparison was set up

PART THREE: RESULTS

9	Test 1: Microsoft	Same rating; the agents found errors and a currency problem
10	Test 2: Coca-Cola	Same rating from opposite sides of one number
11	Test 3: Alphabet	The first disagreement: SELL against HOLD on the same price
12	Every argument the agents made	All 30 challenges and how the judge ruled on each
13	What the tests can and can't tell me	What three tests, two of them agreements, can support
14	What went wrong	My mistakes and the process's, and what changed
15	What it cost	Tokens, dollars and time, and where the cost went

PART FOUR: NOW AND NEXT

16	How I used AI, and conflicts of interest	What Claude did, how I checked it, and what I own
17	Where I am now	Status, the calls log, and what would show the calls were wrong
18	Next steps	What I'll do next, with a timeline to August 2027
19	How to run it yourself	Using the public code, with or without keys
20	What I learned	Lessons for research and for building with agents

After section 20: references, appendix A (build notes) and appendix B (timeline and disclosure).

2 Goal and aims

Why I started

In 2025 I was one of four UVic students on the CFA Institute Research Challenge team, and we wrote a 20-page equity research report on Air Canada. Afterwards I wanted to keep doing that kind of work outside a competition, on companies I chose, at a standard close to what bank research departments publish. I'm also a heavy user of AI tools, and I wanted to find out how far they could take a student working alone, and where they would fall short.

What I set out to do

1. Learn how an equity research report is built by writing real ones, with every number traced back to a company filing.
2. Make the process repeatable, so that any two reports can be read side by side and compared.
3. Test whether splitting the work across AI agents that argue with each other catches things a single line of reasoning misses.
4. Record every call with a date, a price and a target, so each one can be checked against the market a year later.

What would count as success

Aim	Success looks like	Status, September 2026
1. Learn by writing reports	Reports I'm comfortable putting my name on, where every figure has a source	Nine reports finished
2. A repeatable process	The same sections, valuation method and rating rules for every company	In place for both methods
3. Test the agents	The agents catch real errors, and at some point change a rating	Errors caught; the two methods first disagreed on Alphabet in September 2026
4. A checkable record	Calls that beat the market index more often than they miss	Unknown until June 2027

What this paper is not

It isn't investment advice, and it doesn't claim ten agents beat one analyst. Three tests can't show that. It is a record of what I tried, what happened and what I changed because of it.

3 Background and terms

About me

I'm studying commerce at UVic's Gustavson School of Business and graduate in 2027. This term I'm on exchange at SKEMA Business School in France. I sat the CFA Level 1 exam in August 2026 and am waiting on the result. I've done case competitions, a consulting internship and the Research Challenge, but I've never worked in an equity research department, so the standards in this paper come from reading published research, not from having been trained on the job.

My portfolio

I invest a small amount of my own money, in Canadian dollars, through tax-free savings accounts (TFSAs). That matters for this project in two ways. Method 2's risk agent checks every call against what I actually hold, which is how it found the currency problem in section 9. And I own shares in some of the companies I write about, so every report states my position. My current holdings are public on Blossom at blossomsocial.com/users/Vidwaan-Singhania__d7oVAMV5EWqGMPon.

Terms used in this paper

Term	What it means here
DCF	Discounted cash flow: forecast a company's future free cash flow and discount it back to today to estimate what the business is worth
Reverse DCF	The same model run backwards: what growth the current share price assumes
Cost of capital	The discount rate in a DCF, built from interest rates, the market's risk premium and beta
Beta	How much a stock tends to move when the market moves. A beta of 0.5 means about half as much.
Blume adjustment	A standard formula that pulls a measured beta part of the way toward 1.0, the market average
Target and expected return	The price I expect in twelve months, weighted across bull, base and bear scenarios. Expected return is the move to that target plus dividends.
BUY, HOLD, SELL	BUY means an expected return above +15% over twelve months, SELL below -15%, HOLD anything between
Alpha	A stock's return minus the market index's return over the same period
Agent	A separate Claude session given one role, its own instructions and its own files
Tokens	The units AI models are billed in. "Effective tokens" weights each kind by its price, so different runs can be compared.

4 Tools used

Everything the project runs on. I list the model versions because every claim in this paper about cost, about how the debate behaved and about what the judge accepted is a claim about particular models running in a particular harness.

Tool	What it did here
Claude Code	The harness. It runs Claude against the files in my vault, starts the ten agents in a Method 2 run, and keeps the session logs the cost figures in section 15 are counted from.
Claude Opus 5 claude-opus-5	The reasoning in Method 1, and every Method 2 stage from the valuation onward: the valuation agent, the bull, the bear, the judge, the risk agent and the report writer.
Claude Sonnet 5 claude-sonnet-5	The four Method 2 evidence agents: fundamentals, market, news and industry.
Obsidian	The vault of linked notes that holds every report, working file, rule and the calls log.
Python 3.13	The valuation models and the scripts that build each report's PDF, Word file and Excel model, using reportlab, python-docx, openpyxl, matplotlib and pypdf.
SEC EDGAR	Company filings (10-K, 10-Q, 8-K) and the XBRL company facts behind them. Every request has to carry a name and email or EDGAR refuses it.
Alpha Vantage	Prices, analyst estimates and earnings call transcripts, on keys capped at 25 calls a day. A Method 2 run may spend 75 calls and usually spends far fewer: six on Alphabet, one on Coca-Cola.
FRED	Treasury yields for the risk-free rate, series DGS10.
stockanalysis.com	Reference prices, peer multiples and consensus targets, which is what both methods fall back on when the market-data budget is spent.
Headless Chrome	Prints this paper to PDF from the HTML source, and prints one of the Method 1 reports.
PyMuPDF	Rendered the report pages shown in section 8.
Git and GitHub	Publishing the Method 2 code.

Claude is the only tool here that produces analysis. The rest fetch data, do arithmetic I specified, or turn finished text into files. Where a number comes from a model rather than a filing it is marked as an estimate in the report it appears in, and section 16 sets out what I did to check the model's work.

5 What I built on

I haven't done a full literature review. These are the sources I actually worked from, and how my project differs from each.

TradingAgents

Xiao, Sun, Luo and Wang (2024) describe a trading framework modelled on a trading firm. Language-model agents take roles such as fundamental, sentiment and technical analyst, bull and bear researchers debate, a trader decides, and a risk management team monitors exposure. They report backtests in which the framework beat baseline strategies on cumulative return, Sharpe ratio and maximum drawdown. Their open-source code is where I took the idea of separate roles and a debate from.

My project differs in what it produces and how it's judged. TradingAgents makes trading decisions and is tested by backtesting. Method 2 produces a twelve-month BUY, HOLD or SELL call with a written report and a valuation model, and it will be judged on how each call does against the index a year later, with no backtest. I also compare it against a process led by a person on the same companies. I haven't tried to reproduce their backtests.

How professional research is written

My standard came from reading published work: an RBC Capital Markets report on Apple, and the format of the CFA Institute Research Challenge, which I competed in during 2025. I have never seen how a research department trains its analysts or reviews their work, so my checklist is my reading of finished reports, not a description of how they are made.

What might be new here

As far as I know, the unusual part of this project is running a process led by a person and a multi-agent process side by side, on the same stocks at the same price, and logging both so they can be scored against the market later. I may be wrong about that, and I'd like to hear about similar work.

6 Method 1: writing reports on my own

Method 1 is me working through a company with Claude, one section at a time, into the same template every time.

The standard I aimed for

I wrote the first report in mid-June 2026. I kept two documents open beside the work as the bar to meet: an RBC Capital Markets research report on Apple, and my own team's Research Challenge report on Air Canada. From those I wrote a checklist covering tone, citations, how each metric is defined, rating language, valuation method, how risks are laid out, and the disclosures every report has to carry. A report isn't finished until it passes.

Financial statements come from company filings on SEC EDGAR. From July 2 I also used the Alpha Vantage API for prices and estimates. Every figure goes into a working-notes file with its source before it goes into a report, so any number in a finished PDF can be traced back.

What every report contains

#	Section	What it holds
00	Full report	All sections in one read, plus the exported files
01	Company overview	Business model, segments, revenue mix
02	Investment thesis	The main arguments, and where I disagree with consensus
03	Industry and competition	Competitive advantages, named competitors, threats
04	Financial analysis	Growth, margins, cash flow, balance sheet, quality of earnings
05	Valuation	DCF, comparable companies, reverse DCF, sensitivity tables
06	Risks	Specific risks ranked by how much they would move the value
07	Catalysts and ESG	Dated events that could move the shares
08	Scenarios and recommendation	Bull, base and bear cases with probabilities, the target and the rating
09	Appendices	Statistics, assumptions, sources and four disclosure blocks

The valuation always runs the same way. The DCF is the main method and comparable companies are the cross-check. Real costs get charged, including stock-based pay as a cash cost for software companies. The bull, base and bear scenarios get probabilities that add up to one, and those give a probability-weighted twelve-month target. Each report is built into a PDF, a Word file and an Excel model.

The nine companies

By August 16 the reports covered software, AI infrastructure, consumer staples and one copper-gold explorer with no revenue yet.

Company	Date	Price	Call	Target	What decided the call
ServiceNow	Jun 20	\$95.12	BUY	\$138	Down 52% in a year on fears that AI would hurt seat-based pricing, when more than half of new contract value was already usage-based. A DCF charging stock-based pay as cash still gave a \$134 base case.
IBM	Aug 8	\$237.28	HOLD	\$235	A record 25% one-day fall removed the overvaluation without making it cheap. About \$22B of acquisitions against \$43B of free cash flow over 2024–26, so the DCF counts acquisitions as a cost of growth.
IREN	Aug 8	\$41.23	HOLD	\$44	About \$29M of capital per megawatt earning about \$8.3M a year. The first five-year GPU cycle doesn't quite pay back the capital; the return to shareholders comes in the second.
Mogotes Metals	Aug 8	C\$0.64	BUY	C\$1.05	No revenue and no defined resource. Valued on the sum of its parts and on deal prices, including Rio Tinto buying about 5% at C\$0.70, with a stated 25% chance of losing two thirds.

Company	Date	Price	Call	Target	What decided the call
Apple	Aug 8	\$313.33	HOLD	\$270	About \$0.11 of EPS came from a one-off \$2.19B tariff refund, so a record quarter was really a two-cent beat. 1.4 points from the SELL line.
Alphabet	Aug 9	\$354.30	HOLD	\$330	\$6.26 of the quarter's \$9.11 EPS was an unrealised gain on investments. On 2027 estimates the shares trade at 24.0 times earnings, not the 17.8 times a stock screener shows. Refreshed Sep 20 at \$349.54 to a \$340 target after the ad-tech ruling closed the break-up risk.
Coca-Cola	Aug 10	\$87.05	HOLD	\$82	A DCF value of \$67, and a reverse DCF needing 7.88% free cash flow growth for a decade. A \$20.5B IRS tax case, \$4.76 a share, sits outside company guidance. Refreshed Aug 21 at \$91.10.
Microsoft	Aug 11	\$506.06	HOLD	\$500	Reported free cash flow of \$67.0B falls to \$26.8B once unpaid capital spending and lease-funded data centres are counted, against \$48.7B returned to shareholders.
Costco	Aug 16	\$961.10	HOLD	\$923	52.8 times earnings. The reverse DCF needs 15.6% free cash flow growth for ten straight years, against a 9.9% track record.

All dates 2026. Prices are the reference price in each report. Mogotes trades in Canadian dollars on the TSX Venture Exchange.

What I learned writing them

Most of what I learned was about method rather than about any one stock. These lessons became rules, and Method 2 later inherited them.

Reported cash flow can be the wrong number. Microsoft's fiscal 2026 was its best year on record: revenue up 17.8% to \$331.8B and Azure past \$100B. Free cash flow still fell 6.5%, and the real gap was bigger. \$19.8B of capital spending sat unpaid at year end, up from \$6.9B, and lease liabilities rose \$20.4B to pay for data centres that never show up as capital spending. Counting both takes free cash flow from \$67.0B to \$26.8B. Alphabet showed the same pattern two days earlier. I now run that adjustment on every company that spends heavily on physical assets.

So can reported earnings. Alphabet discloses that \$6.26 of its \$9.11 quarterly EPS came from an unrealised gain on investments, and Apple's record quarter included a \$2.19B tariff refund. When a company puts a number on a one-off, I value the shares on the earnings left over and say so in the report.

A DCF tells you what the price assumes. On four of the nine companies my DCF came out far below the share price: Costco at \$529 against \$961, Coca-Cola at \$67 against \$87, Microsoft at \$342 against \$506, and Alphabet about 50% below. On Alphabet, 71% of the value sat in the terminal value, so the answer depended more on the discount rate than on the business. In those reports I show the DCF and reverse DCF as "what the price requires" and build the twelve-month target on earnings multiples instead, because prices don't move to a DCF value within a year. Costco can be an excellent business and still be a HOLD.

Three ratings, and say how close the call is. On August 9 I cut my rating scale to BUY, HOLD and SELL. Apple went from REDUCE to HOLD and Mogotes from SPECULATIVE BUY to BUY, with the risk moved into guidance on position size. Any call near the edge of a band now has to say how near, and name the one assumption that would flip it. The Coca-Cola model prints the three prices where its rating changes: SELL above \$99.03, zero expected return at \$84.18, BUY below \$73.20.

My worst mistakes weren't in the models. On August 9 I rebuilt and re-read every report. Three said I held no position in companies I actually owned, and one had no ownership disclosure at all. Every report now ends with the same four disclosure blocks, and the ownership line is copied from my holdings file.

Where Method 1 fell short

By mid-August I was happy with the reports, but four problems couldn't be fixed by being more careful inside the same process.

- Nothing argued with me. Every assumption was chosen by the same reasoning that later defended it. If a disclosure error could sit in a finished report for weeks, so could a wrong sensitivity table.
- No call would ever be checked. Nine ratings had prices, targets and dates, and nothing in the process would look at them again, so I could only learn whether the reports looked good, not whether they were right.
- Each stock was judged on its own. Method 1 never asked what a call meant for the portfolio holding it, including costs like converting Canadian dollars to US dollars.
- The tooling didn't scale. Every company folder had its own copy of a build script about a thousand lines long, so a fix in one report reached one report.

The question for a second method was whether a structure where agents argue with each other catches things a single process can't, and whether that shows up in the results a year later.

7 Method 2: ten agents

On August 17 I read the code for TradingAgents, an open-source project from Tauric Research (Apache-2.0 licence) that splits stock research across specialist AI agents. I kept its idea of separate roles and rebuilt the process around Method 1's rules.

Five things I changed from TradingAgents

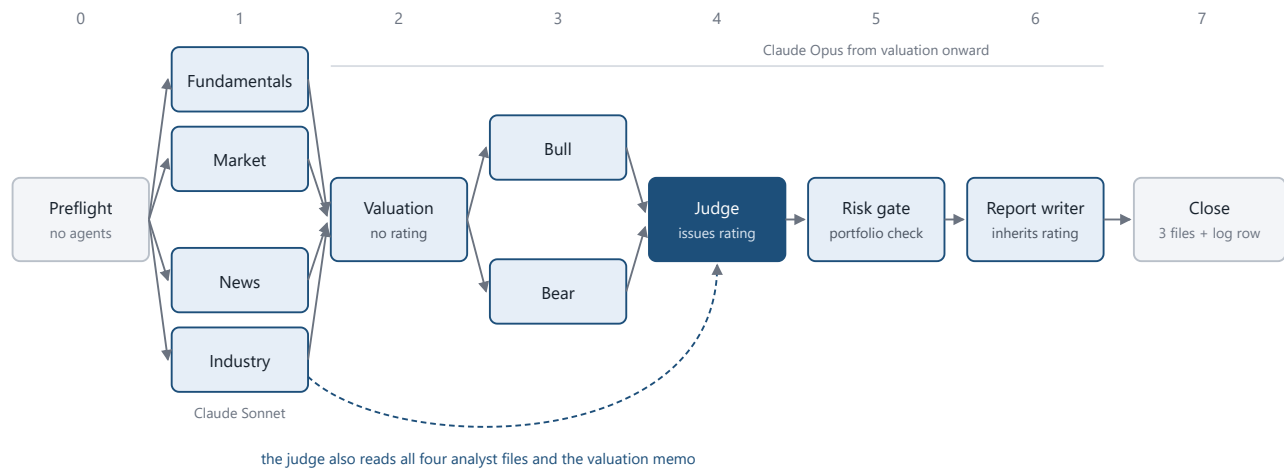
Their design	What I did instead	Why
A five-level rating pulled out of the final text by pattern matching, which defaults to Hold if it finds nothing	Three ratings in a fixed field; any other value stops the run	A parsing failure should show up as an error, not as a call no agent made
The judge reads only the debate	The judge also reads the four evidence files and the valuation memo	Otherwise the more persuasive agent wins, whatever the evidence says
Bull and bear argue in general terms	Both attack a written base case, name the assumptions they dispute and calculate the effect on the target	An argument against a number can be checked
A social-media sentiment analyst	Dropped	Mostly noise for a twelve-month call
Three risk agents with different temperaments debate	One risk agent calculates concentration, correlation and currency cost, and can't change the rating	I wanted risk measured, not argued

The idea I kept exactly as it was is TradingAgents' best one: a log of every call, never edited afterwards, and reviewed later against what the market did.

How a run works

Ten agents in eight stages

Stages run left to right. Stacked boxes run at the same time and can't see each other's work.



Stage	Who	What happens
0 Preflight	No agents; the main session	Sets one reference price for everyone, downloads the SEC filing index and ten years of financial data, extracts the statements, measures beta and records my position in the stock.

Stage	Who	What happens
1 Evidence	Fundamentals, market, news and industry analysts	Four agents work in parallel. Only the market agent may use the market-data API, capped at 75 calls a day.
2 Base case	Valuation agent	Builds the DCF, comparables, three scenarios with probabilities and a sensitivity table. It isn't allowed to give a rating.
3 Debate	Bull and bear	Each attacks the base case with numbers and says what evidence would change its mind. Same model, same time.
4 Judgment	Judge	Gives the rating, records which stage's finding decided it, and says how close the call is to the edge of its band.
5 Risk gate	Risk agent	Checks concentration, correlation with my other holdings, currency cost and entry prices. It can disagree in writing but can't overrule.
6 Report	Report writer	Writes the report in Method 1's section order and copies the rating and target without recalculating them.
7 Close	No agents; the main session	Builds a report PDF, a 14-sheet Excel model and a Word file with every agent's full output, writes the comparison with Method 1 and logs the call.

The rules that matter most

No agent may open my Method 1 report on the same company until the judge has committed to a rating. Without this rule, ten agents would read my conclusion and agree with it.

No number is typed twice. The valuation lives in one Python file that both the Excel model and the PDF read from, and the rating and target live in one decision file. If an assumption is corrected, every output changes when it's rebuilt.

The DCF is always built the same way: free cash flow to the firm, ten forecast years, mid-year discounting, a terminal value tied to growth and return on capital, and a reverse DCF as a check. Two runs that discount differently can't be compared, and comparing is the point.

From Method 1 it kept the three ratings and the rule to say how marginal a call is, ownership stated as a fact, price and quality treated as separate questions, the same report sections and the same checklist.

How the design settled

The first run, on August 17, was on Home Depot using an earlier nine-agent version. Comparing it with earlier work showed three gaps: three years of statements instead of ten, no Porter's five forces table and no football-field valuation chart. By that evening each gap had an owner, a tenth agent had been added so the judge would only judge and a separate agent would write, and the export scripts were shared across runs instead of copied into each one. I later withdrew the Home Depot call. Its row stays in the log marked withdrawn, because a log that quietly drops calls can't be scored honestly.

What a decision record looks like

The judge's decision is saved as a file with fixed fields at the top, followed by its written reasoning. The build scripts read the rating and target from these fields and nowhere else, and a run fails if the rating is anything other than BUY, HOLD or SELL. This is the Coca-Cola record, shortened:

```
ticker: KO
run_date: 2026-08-21
reference_price: 91.10
rating: HOLD
target: 77.45
expected_total_return: -0.1262
conviction: low
marginal: true
flip_assumption: Beta. At the 0.45 I use, the target is USD 77.45 and the
  total return -12.6%. At beta 0.476 the return is -15.0% and the rating
  is SELL; at the valuation memo's 0.56 it is -21.8%.
decided_by: bull
scenarios:
  bull: { probability: 0.25, target: 104.12 }
  base: { probability: 0.50, target: 78.55 }
  bear: { probability: 0.25, target: 48.57 }
models:
  analysts: claude-sonnet-5
  decision: claude-opus-5
alpha_vantage_calls: 1
```

8 How I compared the two methods

First the two methods side by side, then what I tried to hold constant, what I couldn't, and what I measured.

The two methods side by side

	Method 1	Method 2
Who does the reasoning	Me, working through the company with one Claude session	Ten Claude agents with separate roles. I design the process and review what comes out.
Evidence	Gathered in the same session that writes the report	Four analysts gather it in parallel, and none can see the others' work
Checks on the reasoning	My own review against a checklist	A bull and a bear attack the valuation, and a judge rules on the evidence
Valuation	DCF plus comparables; the target usually comes from earnings multiples	A fixed DCF build plus comparables; the judge decides which to rely on
Rating	BUY, HOLD or SELL, written into the report	BUY, HOLD or SELL in a fixed field; anything else stops the run

	Method 1	Method 2
Portfolio check	None	A risk agent checks concentration, correlation and currency cost
Output	PDF, Word file and Excel model	PDF, 14-sheet Excel model, and a Word file with every agent's full output
Cost per report	\$27 to \$35 on the two test companies	\$74 to \$88
Active time per report	0.7 to 1.2 hours	About 2.6 hours
Scored later	Yes, in the calls log	Yes, in the same log

What the two methods produce

Opening pages of the two Coca-Cola reports, both dated August 21, 2026.

EQUITY RESEARCH | Update | Consumer staples — non-alcoholic beverages | August 21, 2026

The Coca-Cola Company

NYSE: KO - USD 91.10 (Aug 21, 2026 close)

HOLD

12-month price target USD 82

A very good quarter at a price that needs everything to go right

My view

The June quarter was the best Coca-Cola has printed in years: organic revenue up 6%, unit case volume up 5% led by India, China, the United States and Brazil, comparable operating margin 35.6% against 34.7%, value share gained in four of five segments, and full-year guidance raised on July 28. None of that is in dispute. **The price is.** At \$91.10 the shares trade at 26.8x 2027 consensus, having run 39% off the \$65.35 low on a 1.4% upward revision to 2027 estimates, so almost the whole move is multiple rather than earnings. Three of the guided nine to ten points of earnings growth are currency; the company's own currency-neutral figure excluding divestitures is 7% to 8%. A discounted cash flow at a 7.16% cost of capital values the shares at \$67. This is the August 21 refresh of a note first written on August 10; nothing has been filed since, consensus for 2027 is unmoved at \$3.53, and the shares are 4.7% higher. I reiterate **HOLD** with a 12-month target of **\$82**, now a -7.6% total return. Accumulate below \$62; trim above \$95.

Key points

The re-rating has already paid for the improvement. From \$65.35 to \$91.10 is a 39% move while consensus 2027 earnings went from \$3.479 to \$3.526, a 1.4% revision, and have not moved since August 10. The reverse discounted cash flow says \$91.10 requires 7.88% free cash flow growth in every year for a decade, against guidance of about 5% organic revenue growth and 7% to 8% currency-neutral earnings growth. The price now asks for better than the top of guidance, forever.

An IRS transfer-pricing case with \$20.5B at stake sits outside every line of guidance. Coca-Cola paid the IRS \$6.0B in September 2024 for the 2007–2025 years and carries a \$514M interest receivable on it. If the Tax Court methodology is upheld and applied to later years the company estimates a further \$14B for 2010–2025. That is \$4.76 a share, 5.5% of the market capitalisation. The appeal was argued at the Eleventh Circuit on June 25, 2026 and a decision is pending. The guided 19.9% tax rate describes the outcome where Coca-Cola wins.

Volume is growing fastest where price per case is lowest. Asia Pacific grew volume 8% and gave back 9% of price/mix on affordability packs in India and China. North America, by contrast, took 4 points of price on 3 points of volume. Consolidated price/mix of +2% is the arithmetic of that mix shift, and it caps growth until those markets premiumise.

Reported margins are noisy and the noise flattens the current year. The 2024 operating margin of 21.2% carried fairlife earn-out remeasurements that reached \$6.173M and were paid in March 2025; its absence is most of the recovery to 28.7% in 2025. The 2025 accounts also carry a \$1.274M impairment on the African bottling operations, booked below the operating line. Comparable operating margin of 35.6% is the cleaner read.

Capital returns are the dividend, not the buyback. The board raised the quarterly dividend to \$0.53 in February 2026, an annualised \$2.12 and the 64th consecutive annual increase, costing about \$9.1B against guided free cash flow of \$12.4B. Net repurchases were \$412M in the first half on a \$374B market capitalisation. Any model that assumes share-count shrinkage is wrong.

Scenario analysis				
	Base	Base	Base	Target
Value	\$09	\$05	\$100	\$60
Return	-30%	-0%	+17%	-3%
Weight	25%	50%	25%	—

Key statistics	
Market cap	\$392.0B
Shares out / diluted	4,303M / 4,313M
52-week range	\$65.35–\$91.87
Dividend / yield	\$2.12 / 2.33%
Dividend increase	64 consecutive years
Buyback yield (H26, net)	0.11%
Cash + securities / debt	\$16.4B / \$43.5B
Net debt	\$27.2B
Trailing P/E	27.4x
Forward P/E (FY2027E)	25.8x
EV / EBITDA	21.2x
FCF yield (2026E)	3.16%
Return on invested capital	18.2%
Beta	0.34
IRS exposure	\$4.76 / share
Consensus / target	Buy / \$84.70

Estimates (FY, December)				
	24A	25A	26E	27E
Rev (\$B)	47.1	47.9	49.7	49.8
Growth	2.9%	1.9%	3.7%	0.2%
Op margin	21.2%	28.7%	—	—
FCF (\$B)	4.7	5.3	12.4	—
FCF ex-fairlife	4.7	11.5	12.4	—
EPS	2.46	3.04	3.30	3.53
Growth	-0.4%	23.6%	8.6%	6.9%
P/E	—	—	27.6x	25.8x

Sources: Coca-Cola Form 8-K Exhibit 99.1 of July 28, 2026 and Form 10-Q for the quarter ended July 3, 2026; Form 10-K for FY2025; consensus across 17 to 24 analysts, and prices and multiples at the August 21, 2026 close; FY2026E and FY2027E revenue and earnings are consensus; analysts; FY2026E free cash flow is company guidance. The FCF ex-fairlife line adds back the \$6.173M contingent consideration payment made in March 2025, which is why headline 2026 free cash flow appears to more than double.

Vidwan Singhania - Independent equity research for educational purposes - not investment advice

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What I controlled, and what I didn't

Condition	Microsoft	Coca-Cola	Alphabet
Same reference price	No: \$506.06 against \$480.35, six days apart	Yes: \$91.10 for both	Yes: \$349.54 for both
Same filings available	Yes	Yes; no new filing between the two	Yes
Method 2's agents kept away from the Method 1 report	Yes	Yes	Yes
Me kept away from either answer	No	No	Partly: Method 2 ran first, but Method 1's August call existed
Same rating bands	Yes	Yes	Yes
Same DCF construction	No; Method 2 uses its own fixed build	No	No
Same AI (Claude) doing the reasoning	Yes	Yes	Yes

What I measured

For each company: whether the ratings match, how far apart the targets are, which inputs differ and by how much, every error the agents found and whether the judge accepted it, which stage the judge said decided the call, and the tokens, dollars and time each method used.

How the calls will be scored

Each call is reviewed twelve months after it was made. Its return is the price change plus dividends. Its alpha is that return minus the total return of its benchmark over the same dates: the S&P 500 (through the SPY fund) for US stocks, and the S&P/TSX Capped Composite (through the XIC.TO fund) for Mogotes. A BUY is right if the stock beat its benchmark and a SELL is right if it lagged. A HOLD is right if the stock ended roughly inside the range the call implied, which is the hardest of the three to judge, so I plan to judge it strictly. Each scored call gets a short written reflection, and those reflections are fed to the judge on later runs. Calls that haven't been scored are never fed forward.

9 Test 1: Microsoft

The first real test ran on August 17, six days after my Method 1 report. Microsoft's shares fell 5.1% in between, so some of the difference is the price move and not the method.

	Method 1	Method 2
Rating	HOLD	HOLD
Reference price	\$506.06, Aug 10	\$480.35, Aug 17
Twelve-month target	\$500	\$487.18
Conviction	Moderate	Moderate
DCF value	\$342	\$279.54
Cost of capital	9.89%	9.22%
Scenario weights, bull / base / bear	25 / 50 / 25	28 / 50 / 22
What carried the call	Valuation	The market analyst's history of multiples and estimate revisions

Method 1's reference price is the August 10 close. The report was finished and dated August 11, which is the date the calls log uses.

Same rating, same conviction, targets 2.6% apart. Both methods found the DCF well below the share price and both chose not to base the twelve-month target on it. Both landed on capital spending as the thing that matters. Put simply, ten agents reproduced what I had reached a week earlier, at about \$74 of tokens against \$35 for my Method 1 session (section 15). What made the run useful were three findings Method 1 had no stage to produce.

The bear found mistakes, including one that helped the bull

The bear agent rebuilt the valuation model in Python and matched the base case to the cent before changing anything. It then found that one cell of the sensitivity table had applied only half of a two-part change. Corrected, the top of the DCF range rose from \$311.12 to \$323.31, so it overlapped the scenario range by \$7.79. That overturned the valuation memo's main claim that the two ranges didn't touch, and it raised the valuation, which is the opposite of what the bear was arguing. It also found that the fiscal 2028 EPS figure implied a share count that appeared nowhere else in the model. The judge accepted both corrections.

The currency cost was bigger than the expected return

I buy US stocks with Canadian dollars. Measured on my own conversions, changing money there and back costs about 3.77%, and the expected total return on the HOLD was +2.23%. Seen from the sell side: bought at the \$480.35 reference price, Microsoft would have to reach \$494.58 within a year just to break even, \$7.40 above the target. Seen from the buy side: with a \$487.18 target and about \$3.89 of dividends, a purchase only covers the currency cost below roughly \$473. My Method 1 report said to buy more anywhere below \$500, so on my account that advice lost money at every price from about \$473 to \$500, even if the call turned out exactly right. Method 1 had no step that would have checked.

A disagreement on the discount rate

Method 2 built a 9.22% cost of capital from a 4.72% ten-year Treasury yield, a 4.23% market risk premium and a beta of 1.10. The bear argued for 9.84% on a beta of 1.25, and the judge rejected it. Method 1, working

alone a week earlier, had used 9.89%, the bear's side of the argument. The two methods are 67 basis points apart on this and I haven't settled which is right.

What decided the call

The judge moved the scenario weights from 25/50/25 to 28/50/22 because of one piece of evidence the valuation memo had recorded but not used: 39 upward revisions to analysts' EPS estimates against 13 downward in the month after results. Neither side's preferred weights could move the rating out of HOLD, because the call depended on the scenario targets, not the weights.

10 Test 2: Coca-Cola

Coca-Cola on August 21 was the cleaner test. I refreshed the Method 1 report that evening using the same \$91.10 closing price Method 2 used, so the gap between the two calls can't be explained by price movement, time passing or a new filing.

The order matters, so I checked it against the file timestamps. The refreshed Method 1 valuation and recommendation were saved at 9:17 pm Pacific. Method 2 started a minute later, and the rest of the refresh (the risks section and the rebuilt files) was done by 9:25 pm, before any Method 2 agent had written anything. No Method 1 file changed after that. API limits then stretched the Method 2 run to August 23. So Method 1 couldn't have copied Method 2, but I did know Method 1's answer before Method 2 started, which I come back to in section 13.

	Method 1	Method 2
Rating	HOLD	HOLD
Twelve-month target	\$82	\$77.45
Expected total return	-7.6%	-12.6%
Distance to the SELL line	7 points	2.4 points
Beta used	0.558	0.45
Where the target comes from	24x earnings multiple; its own \$67 DCF not used	The DCF, at \$78.55 base
\$20.5B IRS tax case	Bear case only	Probability-weighted in every scenario
Scenario range	\$58.89 to \$99.90	\$48.57 to \$104.12

Same rating, from opposite sides of one number

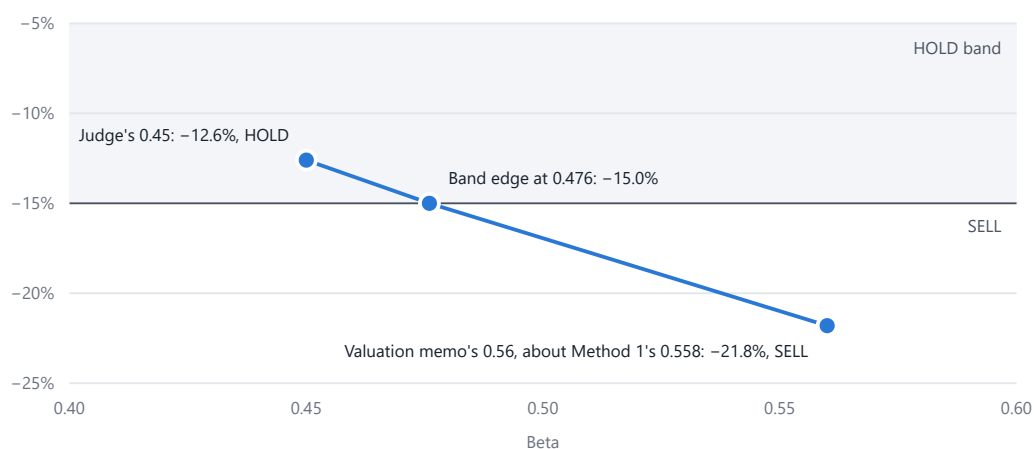
Coca-Cola's measured beta was 0.34. Method 1 applied the standard Blume adjustment, which pulls a measured beta toward the market average of 1.00, and got 0.558. Method 2's valuation agent reached 0.56 in a slightly different way. Then the bull agent checked how the memo defended that number. The memo said

0.55 was the highest beta in Coca-Cola's peer group and belonged to Monster Beverage. The market analyst's own file put Monster at 0.52. The 0.55 belonged to Coca-Cola Consolidated, a bottler the same file had excluded as not comparable.

The judge took the point further. The Blume adjustment pulls toward 1.00, the average for any stock, but every beverage company in the evidence (bottlers aside) measured between 0.34 and 0.52. For a company with a clear peer group, pulling toward the peer average is a better estimate. The judge used 0.45, a little above the peer average, to allow for the chance that the five-year measurement window was unusual for the whole sector.

Coca-Cola: Method 2's expected total return at three betas

Method 2's model, \$91.10 reference price, dividends included. Below -15% the rating is SELL.



Source: Method 2 Coca-Cola decision record, August 21, 2026. The line joins the three published points; values in between were not calculated.

The beta change was worth \$8.39 of the \$77.45 target. Every other correction the judge made added up to \$0.46. Put Method 1's beta into Method 2's model and Method 2 becomes a SELL at -21.8%. Both methods came to the same view of the business, a very good company at a price that needs everything to go right, and split on a number that neither had measured more than once. That went straight into the process: beta is now measured before any agent starts, over several time windows and with standard errors.

Two more things Method 2 caught

The bear found that the valuation memo subtracted Coca-Cola's \$6,514M IRS deposit and interest from a starting figure that had never included it. Fixing the double count raised the target by \$0.54, again helping the side the bear was arguing against. The risk agent then checked a claim in the judge's own decision record, that the currency cost was bigger than any return the entry prices could capture, found it wrong on my measured conversion costs, and left the disagreement on the record.

Where Method 1 did better

Method 1's Coca-Cola report had much more detail on each business segment. It also made a point that Method 2's four analysts each saw part of but none put together: nothing about the business changed between August 10 and August 21, and the shares rose 4.7% on the multiple alone. Method 1 was also clearer about what would change the call, naming a price and two specific conditions. On Microsoft, Method 1 listed nine risks where Method 2's template allows five.

Which stage decided

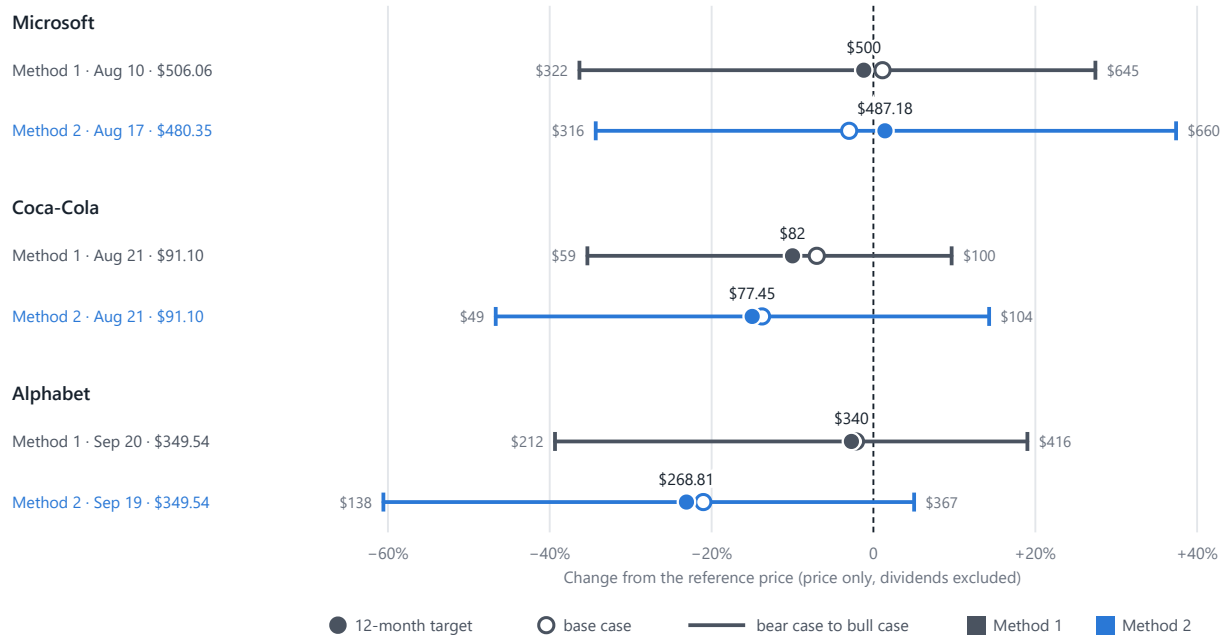
Every Method 2 decision record names the stage whose finding carried the call. On Microsoft it was the market analyst. On Coca-Cola it was the bull, the first time a debate agent rather than an evidence or valuation stage set the answer. On Alphabet it was the market analyst again, and that run is the one where the two methods split. Three runs prove nothing, but after a dozen this field should show whether the debate does real work.

All three tests side by side

The chart puts the six calls on one scale: each method's bear case, base case, bull case and target as a change from the price it started from. On Microsoft the two methods are within a few points of each other everywhere except the bull case. On Coca-Cola, Method 2 is below Method 1 at every point except the bull case. On Alphabet the whole Method 2 row sits below the Method 1 row, which is what a rating disagreement looks like on this scale.

Scenario ranges and targets, both methods, all three companies

Each row runs from the bear case to the bull case. Dollar labels are the scenario values.



Sources: Method 1 Microsoft report section 08, Coca-Cola and Alphabet comparison files; Method 2 decision records of August 17 and 21 and September 19, 2026. Method 1's Microsoft scenarios are on fiscal 2027 earnings and Method 2's on fiscal 2028, so those two rows aren't built on identical inputs.

11 Test 3: Alphabet

The third test, on September 19, is the first time the two methods disagreed. Method 2 called Alphabet a SELL. Method 1, refreshed the next day off the same closing price, called it a HOLD.

	Method 1	Method 2
Rating	HOLD	SELL
Reference price	\$349.54, September 18 close	\$349.54, September 18 close
Twelve-month target	\$340	\$268.81
Expected total return	–2.5%	–22.8%
Conviction	Moderate	Moderate
Scenario weights, bull / base / bear	32 / 48 / 20	30 / 45 / 25
Base case	\$342: 23.0x on \$14.87 of earnings	\$276.02: 21.0x on \$13.14
Where the earnings figure comes from	Street consensus for 2027	Built from the fundamentals agent's depreciation schedule
What carried the call	The refresh: a closed antitrust risk and higher consensus earnings	The market analyst's finding on core earnings

The order, again

Method 2 ran first this time. Its call was committed on September 19 and the Method 1 refresh was written the next day, off the September 18 close both used. That is closer to a blind test than the first two, but it isn't one: Method 1 had published a HOLD on Alphabet in August with a \$330 target, and I knew it.

Where the two agree

Both found the same thing about Alphabet's headline earnings, separately. Two thirds of the June quarter's \$9.11 of EPS was an unrealised gain on equity investments the company itself disclosed, so the business earned about \$2.85. Method 2's market analyst reached that in the evidence stage, before any agent could have seen my report. Both also treated the ad-tech antitrust remedies decision of September 2, which rejected a forced sale of the AdX exchange and ordered changes in behaviour instead, as closing the break-up risk rather than leaving it open. My own August report had the date wrong, reading the September 16 unsealing of the opinion as the decision itself; the agents' news file had it right, and I corrected my node. And both built the twelve-month target from earnings multiples rather than from the DCF, with bull-case multiples landing on the same 26.0x.

Where they differ

Two gaps compound into a rating disagreement. The first is the earnings estimate. Method 1 takes consensus for 2027, \$14.87 a share. Method 2 builds its own, \$13.14, from a depreciation schedule in which servers on six-year lives and structures on twenty push depreciation from 5.2% of revenue today into double digits, which holds the margin flat instead of letting it expand. That is 11.6% lower before any multiple is applied. The second gap is the multiple: 23.0x against 21.0x, where Method 2's judge argued that the worst cash conversion and the highest capital intensity in its peer group earn a discount to the peer average rather than a mid-pack multiple. Method 2 also moved three points of probability from its bull case to its bear case.

Method 2's call is not marginal. It sits 7.8 points inside the SELL band, and on its own scenario weights it only reaches HOLD at a base multiple of 25.6x, which is above the peer average and close to its own bull case.

What the debate did

Both attacking agents rebuilt the valuation and reproduced it to the cent, \$190.98 against \$190.99, so nothing in the ruling turned on arithmetic. They then found errors in opposite directions, which the judge read as a sign the model wasn't tilted one way. The bull found that a revenue cut had been applied to the wrong year, worth \$6.80 a share, and the judge adopted it. The bear found that the model held the share count flat while the company had stopped buying back stock, and that the terminal year charged less depreciation than the model's own check implied; both were adopted in part.

The change I made after Coca-Cola paid off here. The bear argued for a forward beta of 1.40. Because beta is now measured before any agent starts, the judge could answer with four regressions: 1.185, 1.203 and 1.215 over two, three and five years, with the standard error falling to 0.097, against an independently quoted 1.22. The bear's figure was an assertion, and the measurement settled it. On Coca-Cola, a month earlier, nobody had a measurement to point at.

What would settle the disagreement

Both processes flagged the same number without seeing each other's flag: Alphabet's capital spending guidance for 2027, due with fourth-quarter results in late January or early February. Method 2 put a threshold on it. Guidance above \$280 billion supports its reading of capital intensity; at or below \$240 billion that reading breaks, and with it the earnings path that produced the SELL. Third-quarter results in late October are the earlier, partial check.

12 Every argument the agents made

Sections 9 to 11 cover the findings that mattered most. This section lists every substantive claim the bull and bear made against the valuation, and what the judge did with it, so the rejected ones are on the record too.

Coca-Cola

Claim	Raised by	Judge's ruling	Effect on target
The memo's defence of a 0.56 beta rests on a peer beta taken from the wrong company	Bull	Accepted. Beta moved from 0.56 to 0.45.	+\$8.39
The IRS deposit is subtracted from a starting figure that never included it	Bear	Accepted. The bear's own case got weaker.	+\$0.54
The tax rate should change with the chance of losing the IRS case	Both	Accepted. The rate now runs from 20.28% to 22.75% across scenarios.	+\$0.03

Claim	Raised by	Judge's ruling	Effect on target
An unpriced charge for the fairlife production suspension	Bear	Accepted at \$400M, because a one-sided risk shouldn't be carried at zero.	–\$0.09
A 1.0-point cost allowance was described but never deducted, worth \$1.82 a share	Bear	Rejected. The memo's sentence was wrong but its number was right.	none
The chance of losing the IRS case should be 20% (bull) or 55% (bear)	Both	Rejected. The memo's 35% stayed.	none
Terminal growth should be 2.0%, not 3.0%, worth \$6.74 a share	Bear	Rejected, and the judge wrote that this was the rejection most likely to look wrong in a year.	none
Scenario weights should move toward the bull or the bear	Both	Rejected. The two arguments cancelled out, so 25/50/25 stayed.	none
The comparable-company check is circular	Bull	Recorded but left unresolved on purpose.	none

The judge's target ended \$8.85 above the valuation memo's \$68.60: \$8.39 from beta and \$0.46 from the other accepted corrections together. Source: Coca-Cola decision record, section 4.

Microsoft

Claim	Raised by	Judge's ruling
One cell of the sensitivity table applied half its stated change	Bear	Confirmed. The top of the DCF range rose from \$311.12 to \$323.31. No effect on the target.
Fiscal 2028 EPS implies a share count used nowhere else	Bear	Confirmed. Base EPS corrected from \$21.28 to \$21.18.
The bull scenario sits above consensus EPS while claiming to use it	Bear	Accepted. Bull EPS set to the \$23.57 consensus.
39 upward estimate revisions against 13 downward were recorded but not used	Bull	Accepted. Weights moved from 25/50/25 to 28/50/22.
The memo's finance-lease split contradicts its own explanation	Bull	Accepted as a defect in a line that doesn't enter the valuation.
A rule of thumb for capital spending leaves out replacement cost	Bull	Accepted; the rule won't be reused.
Capital spending and depreciation should be lower after a lease reclassification	Bull	Rejected. Company guidance already included the reclassification.
Capital spending should stay flat at \$275B from fiscal 2029 to 2036, worth –\$12.08 a share	Bear	Not adopted, but recorded as the stronger of the two capital-spending attacks.
Anchoring the target to past multiples reproduces any mispricing	Bear	Correct as logic, rejected as a reason to move a twelve-month target.
A base case with 12% growth and \$15B more depreciation	Bear	Rejected as a bear case presented as a base case.

Alphabet

Claim	Raised by	Judge's ruling
A revenue cut was applied to the wrong year	Bull	Accepted, worth \$6.80 a share.
The purchase-commitments figure is not all capital spending, so it doesn't prove permanent capital intensity	Bull	Called the best point in either file, but the 24% terminal capital intensity stayed.
The discount rate mixes a January risk premium with a September bond yield	Bull	Called fair, then offset by the bear's argument that beta should be higher. The 9.90% stayed.
A lower discount rate, worth \$36 a share	Bull	Rejected. It assumes the required return stayed fixed while yields rose.
Terminal capital intensity of 18%, not 24%	Bull	Rejected. Replacing servers alone needs more than that.
Terminal return on capital of 25%	Bull	Rejected as measured on a much less capital-heavy business.
As Cloud grows, the mix pulls the margin down; the model expands it instead	Bear	Partly accepted. Margin held at 44.0% rather than the bear's 42.5%.
The share count can't stay flat while buybacks have stopped	Bear	Accepted. 12,700M shares rather than 12,330M.
The terminal year charges less depreciation than the model's own check implies	Bear	Accepted, worth about \$10 a share.
A forward beta of 1.40	Bear	Rejected. Preflight measured 1.19 to 1.22 across three windows with a standard error of 0.097.
Terminal growth of 2.50%, not 3.50%	Bear	Rejected as charging the same effect twice.
A base exit multiple of 15.0x	Bear	Rejected as below anything the shares have traded at in five years.

Across the three runs the judge accepted 16 of 30 claims in whole or in part. The accepted ones were mostly arithmetic and sourcing errors, which can be checked against a file. The rejected ones were mostly forecasts about the future, which can't. That split seems right to me, but it also means the debate is better at auditing a model than at changing a view. On Alphabet the rating came from the evidence stage rather than from the debate: the market analyst's finding about core earnings is what the judge said carried the call.

13 What the tests can and can't tell me

Two agreements and one disagreement, from three tests. Both parts are weaker evidence than they look, for reasons I should be upfront about.

The two methods aren't independent

I built both. Method 2 inherits Method 1's rating bands, its rule that quality and price are separate questions, its report sections and its checklist, and I wrote the agents' instructions after writing nine Method 1 reports. Claude does the reasoning in both. The anchoring rule stops an agent from reading my conclusion, but it can't stop my conclusions from shaping the instructions the agents follow. On Coca-Cola I already knew the Method 1 answer when I started the run, and on Alphabet I knew the August call even though the refresh came after. So agreement mostly shows that Method 2 reproduces my own method. The more useful evidence is where the two disagreed or where Method 2 caught something: the beta, the discount rate, the arithmetic errors and the currency check.

Other limits

- Three tests is a small sample, and five of the six calls were HOLDs. No test has yet checked whether the two methods agree on a BUY.
- The one disagreement doesn't show that either method is right. It shows that a bottom-up earnings estimate and a lower multiple can move a call by twenty points of expected return.
- No call has been scored yet. Until June 2027 neither method has a track record, and both are opinions with footnotes.
- An experienced analyst might disagree with some of the judge's decisions, such as using a beta of 0.45, and I don't yet have the experience to be sure those decisions were right.
- The cost comparison in section 15 slightly undercounts Method 1, because it only includes the session that wrote each report.

A fairer test would take a company I have no view on, seal Method 2's call first, and only then write the Method 1 report. That is the first item in section 18.

14 What went wrong

A list of my own mistakes and the process's failures, with how each was found and what changed.

What went wrong	How it was found	What changed
Three reports said I held no position in companies I owned; one had no ownership disclosure	My re-read of every report on August 9	Four fixed disclosure blocks; ownership copied from the holdings file
Two months of reports used Helvetica instead of the intended serif font	A later check of the finished PDFs	The build now prints the font it actually used
The first export pass on IREN deleted a paragraph I wanted to keep	Comparing the new PDF with the previous one	Every rebuild is now compared with the last version
My Method 1 advice to buy Microsoft below \$500 lost money after currency costs	Method 2's risk agent	The risk check is now part of every Method 2 run

What went wrong	How it was found	What changed
On the first Microsoft pass the market agent made no data calls	Reading the run log	Each run now passes the agent a table of which key is which
Agents spent a third of the Microsoft run searching for filing addresses	Token audit	Filing addresses and financial data are fetched once, before any agent starts
The valuation agent defended its beta with a figure from the wrong company, an error it copied from the market analyst's file	The bull agent	Beta is now measured before any agent starts
The run file told the risk agent my US allocation was below target, using a figure that had been revised the same day	The risk agent	Flagged in the Microsoft run; the holdings note needed correcting
The Coca-Cola judge claimed the currency cost exceeded any return its entry prices could capture	The risk agent, against my measured conversion costs	Left on the record as a disagreement
API limits killed three stages of the Coca-Cola run; two restarted from zero	The run stopping	Agents now save each section as they finish it
The Home Depot run used an earlier version of the process	Comparing it with later runs	Call withdrawn; its row stays in the log
The first version of this paper misstated the Microsoft currency example	A review of the paper on September 17	Rewritten in section 9
My token audit counted many responses twice, overstating the Coca-Cola run by almost double	Recounting the session logs on September 17	Corrected in section 15

The pattern I take from this list is that most of my mistakes were in routine steps I didn't think needed checking. The agents caught mistakes in their own work as well as in mine, but several of the problems above were in Method 2 itself.

15 What it cost

After each run I went through where the tokens went. The answer surprised me, and it changed the process more than anything I learned about a stock.

Cost per report

Company	Work	Effective tokens	At API prices	Active time	Elapsed
Microsoft	Method 1 report, Aug 11	6.9M	\$35	1.2 h	18 h
Microsoft	Method 2 run, Aug 17–18	18.8M	\$74	2.6 h	12 h
Coca-Cola	Method 1 report, Aug 10	5.3M	\$27	0.7 h	19 h
Coca-Cola	Method 1 refresh, Aug 21	2.3M	\$11	0.3 h	0.3 h
Coca-Cola	Method 2 run, Aug 21–23	21.2M	\$88	2.6 h	34 h
Alphabet	Method 1 refresh, Sep 20	6.8M	\$34	0.5 h	7 h
Alphabet	Method 2 run, Sep 19–20	21.5M	\$71	2.3 h	15 h

Counted from the session logs on September 17, 2026, counting each API response once. Effective tokens weight each kind of token by its price relative to fresh input: cache writes 2x, cache reads 0.1x, output 5x. Dollars use API list prices of \$5 per million effective tokens for Claude Opus and \$2 for Claude Sonnet. Active time adds up the gaps between logged events and ignores any gap over five minutes. Method 1 rows cover only the session that wrote each report.

I ran everything on a Claude subscription, so the dollar column is what the work would cost on the API, not what I paid. On these numbers a Method 2 run costs two to three and a half times a Method 1 report and takes two to four times the active model time. Most of the Coca-Cola run's 34 hours was waiting: API limits stopped three stages, and the run picked up again the next day. The Alphabet run cost slightly less than the earlier two despite covering more ground, because the statement extraction and the preflight work took jobs away from the agents. For that extra cost Method 2 found the currency problem, the arithmetic errors and the beta issue on the first two companies. On the third it produced a different rating.

Microsoft: agents searching instead of reading

On the first full run, the fundamentals and news agents made 66 and 90 tool calls, most of them hunting for the web addresses of SEC filings rather than reading the filings. Gathering evidence was more than a third of the run's cost. The fix was to look up the company's SEC identifier, filing list and financial data once, before any agent starts, and hand them over.

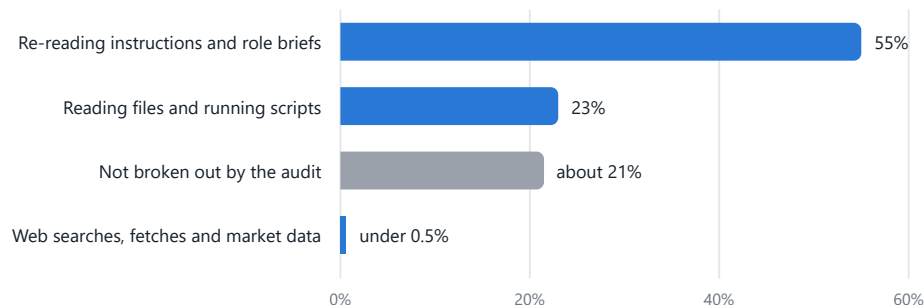
Coca-Cola: agents carrying their instructions

The Coca-Cola run came to about 21 million effective tokens, 85% of it inside the agents rather than the main session. My first audit on August 23 said 39.6 million, and it was wrong. The session log writes one line for each part of a response, and my audit counted every line, so responses with two parts were counted twice. The shares in the chart below come from that first audit. The double count ran through every category, and

the one share I could recheck, the two stages that failed and restarted, came to 8.6% on the recount against 8.5% in the audit.

Coca-Cola run: share of token cost by activity

August 21, 2026 run, about 21 million effective tokens on the recount



Source: token audit of the Coca-Cola run, August 23, 2026. It overstated the total but not the shares (see text). The grey bar is what's left after the three categories the audit measured.

One agent, the fundamentals analyst, spent 2.5 million tokens across 77 tool calls working through a 4.8 MB file of financial data to rebuild statements the data already fully determines. The reason is mechanical. Every time an agent uses a tool, everything earlier in its conversation gets sent again, so cost grows with the length of the conversation times the number of steps. What an agent carries on every step costs far more than what it looks up once.

Three changes I made

Change	What it does	Why
Preflight builds the statements	A script extracts ten years of income statement, balance sheet and cash flow lines before any agent starts. The fundamentals agent checks the rows instead of building them.	On saved Coca-Cola data it matched the hand-built figures on every line checked except 2016 diluted EPS, two cents off because that year was later restated.
Agents save as they go	Every agent writes each section to its file as soon as it finishes it.	API limits killed three stages of the Coca-Cola run. The two holding work in memory restarted from zero, about 1.8 million tokens; the one saving as it went picked up where it stopped.
Beta measured in preflight	Weekly regressions over one, two, three and five years, with standard errors, for the company and its peers.	The Coca-Cola call came down to 0.34, 0.45 or 0.56, argued over by four agents and a judge from one five-year estimate.

The third change is about the quality of the call as much as its cost. None of the three removes an agent or lets the judge skip the evidence.

All three ran for the first time in the Alphabet run on September 19. The statement extractor pulled ten years of figures before any agent started, and the beta regression did the job it was added for: when the bear agent

asserted a forward beta of 1.40, the judge could answer with four measurements and their standard errors instead of an argument.

16 How I used AI, and conflicts of interest

What Claude did and what I did

Claude did most of the reading, calculating and drafting in both methods. I chose the companies, set the standard and the rules, designed both processes, reviewed the outputs, and decided what to change after each run. This paper was also drafted with Claude and edited by me. Everything in it comes from files in my project: the reports, the run records, the calls log and the session logs.

How I tried to keep the AI honest

- Every figure in a report has to be traced to a source in a working-notes file before it's used.
- Financial statements come from SEC filings, not from the model's memory.
- In both Method 2 runs the bear and the judge rebuilt the valuation model from scratch before arguing with it (on Coca-Cola the bull did too), and matched the published base case to within a few cents.
- The judge checks every claim against the evidence files, not against how persuasive the argument sounds.
- The rating is a fixed field, so a formatting mistake stops the run instead of producing a call.

None of this guarantees the numbers are right. It makes errors easier to find, which is how the ones in section 14 were found.

Conflicts of interest

I own shares in Microsoft and Coca-Cola, and in some of the other companies in section 6. Every report states my position as a fact, copied from my holdings file, and my current holdings are public on Blossom. Method 2's agents are given my position so the risk agent can check the call against my portfolio. I haven't found a way to stop owning a stock from influencing how I write about it, other than disclosing it and logging every call so it gets scored.

Data

Everything used is public. SEC EDGAR requires a name and email with each request, which the scripts send. Market data came from free Alpha Vantage keys within their daily limits, from FRED, and from public price pages.

17 Where I am now

As of September 20, 2026.

Part of the project	Status
Method 1 reports	Nine finished. Coca-Cola was refreshed on August 21 and Alphabet on September 20; the other seven still use their August prices.
Method 2 runs	Three live (Microsoft, Coca-Cola and Alphabet) and one withdrawn (Home Depot). The most recent closed on September 20.
Changes from the cost audit	All three ran in full for the first time on Alphabet, and the beta measurement settled an argument in that run.
Calls log	Fourteen rows: thirteen open, one withdrawn, none scored. The first review is due June 20, 2027.
Code	Public on GitHub under the MIT licence, with a full worked example on an invented company.
Calls log gap, closed	The Alphabet Method 1 refresh (HOLD, \$340) had no row of its own until September 20. It has one now, and the August call keeps its row and its own review date. A refresh that moves the target is a new call.
My portfolio	Current holdings public on Blossom at blossomsocial.com/users/Vidwaan-Singhania__d7oVAMV5EWqGMPon .
This paper	Revised on September 17 and 18, and updated on September 20 with the Alphabet test.

The calls log

Every call from both methods is recorded with the price on the day, the target, the benchmark index and a review date twelve months out.

Ticker	Method	Called	Call	Price	Target	Benchmark	Review due	Status
NOW	1	2026-06-20	BUY	\$95.12	\$138	SPY	2027-06-20	open
IBM	1	2026-08-08	HOLD	\$237.28	\$235	SPY	2027-08-08	open
IREN	1	2026-08-08	HOLD	\$41.23	\$44	SPY	2027-08-08	open
MOG	1	2026-08-08	BUY	C\$0.64	C\$1.05	XIC.TO	2027-08-08	open
AAPL	1	2026-08-08	HOLD	\$313.33	\$270	SPY	2027-08-08	open
GOOGL	1	2026-08-09	HOLD	\$354.30	\$330	SPY	2027-08-09	open
KO	1	2026-08-10	HOLD	\$87.05	\$82	SPY	2027-08-10	open
MSFT	1	2026-08-11	HOLD	\$506.06	\$500	SPY	2027-08-11	open
COST	1	2026-08-16	HOLD	\$961.10	\$923	SPY	2027-08-16	open
HD	2	2026-08-17	HOLD	\$338.86	\$303.75	SPY	—	withdrawn
MSFT	2	2026-08-17	HOLD	\$480.35	\$487.18	SPY	2027-08-17	open

Ticker	Method	Called	Call	Price	Target	Benchmark	Review due	Status
KO	2	2026-08-21	HOLD	\$91.10	\$77.45	SPY	2027-08-21	open
GOOGL	2	2026-09-19	SELL	\$349.54	\$268.81	SPY	2027-09-19	open
GOOGL	1	2026-09-20	HOLD	\$349.54	\$340	SPY	2027-09-20	open

Each call will be scored on alpha, its return minus the index's return, not on raw return. A HOLD on a stock that fell 20% while the index fell 2% was a bad call. A HOLD on a stock that fell 20% while the index fell 25% was a good one. Once calls are scored, the judge will get the scored rows for that company, plus the three most recent scored rows on any company, before each new run.

What would show these calls were wrong

Some of the calls can be checked before their review dates, because they depend on things that will be reported sooner.

Call	What to watch	What it would mean
Microsoft, both methods	Capital spending in fiscal 2027 first-quarter results, around late October 2026	The biggest input in the model is an estimate with no consensus behind it; this is the first check
Coca-Cola, Method 2	Fiscal 2026 comparable operating margin	Below 31% means the bear was right about margins and the judge was wrong
Coca-Cola, both methods	The Eleventh Circuit ruling on the \$20.5B IRS case	The largest single swing in every scenario
Coca-Cola, Method 1	A share price above \$99	On Method 1's own numbers the call becomes SELL without any change in the business
Coca-Cola, Method 2	A properly measured beta above 0.476	The call becomes SELL
Alphabet, both methods	Capital spending guidance for 2027, due with fourth-quarter results in late January or early February	Above \$280B supports Method 2's reading and its SELL; at or below \$240B breaks it
Alphabet, both methods	Third-quarter results in late October 2026	The first full quarter at the raised spending rate, and a read on whether free cash flow turns positive
Alphabet, Method 2	The market paying about 25.6 times its 2027 earnings estimate	The call returns to HOLD

Questions I still can't answer

Question	What would answer it	Where it stands
Do the two methods disagree often, and who is right when they do?	More runs, and scored results a year later	Two agreements, one disagreement (Alphabet); none scored
Which stage does the real work?	The deciding stage recorded in each run, across a dozen runs. If the evidence stages always decide, the debate can go.	Market analyst twice, bull once
Is either method any good?	Scored rows in the calls log	First review June 20, 2027

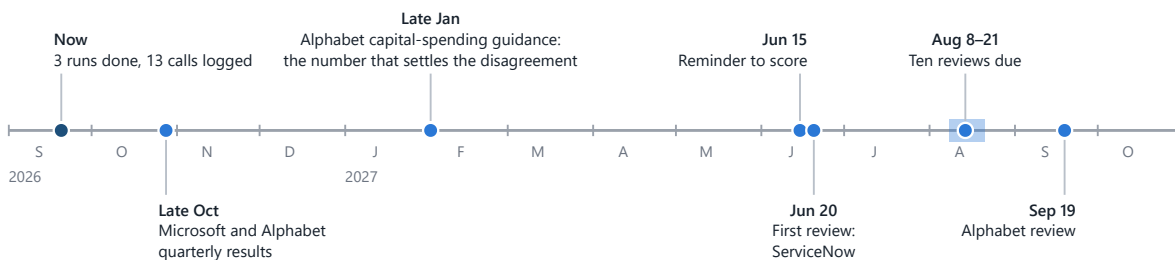
18 Next steps

In the order I plan to do them. The dates are fixed only where the calls log sets them.

The next twelve months

Dated checks and reviews, September 2026 to November 2027.

Not yet dated: the blind test, more runs on covered companies, and cutting the cost of a run.



#	Step	Why	When
1	Run a blind test	Pick a company I have no view on, run Method 2 and seal its call, then write the Method 1 report. Alphabet came close, since Method 2 ran first, but an August call already existed.	Next run
2	Run Method 2 on more companies Method 1 already covers	Three tests isn't enough to say how often the methods disagree, or which one is right when they do.	After step 1
3	Cut the cost of a run	More than half the cost is agents re-reading their instructions, so shorter role instructions should save the most. I'll measure against the \$71 to \$88 in section 15.	Alongside steps 1 and 2
4	Watch the Alphabet disagreement resolve	Capital spending guidance in late January or early February is the number both methods named as decisive. Whichever way it lands, one of the two readings breaks.	Late January 2027

#	Step	Why	When
5	Track which stage decides	After about a dozen runs, cut the debate if it never decides a call, or keep it if it does. So far the evidence stage has decided twice and the debate once.	Once there are twelve runs
6	Set a reminder to score calls	Nothing in the process scores a call automatically. If nobody runs the review, the log turns into a list of opinions that were never checked.	Before June 15, 2027
7	Score calls as they come due	ServiceNow is first. Most of the others come due in August 2027, and Alphabet's Method 2 call in September 2027.	From June 20, 2027
8	Write a follow-up paper	Report the scored results, including the calls that went wrong.	After August 2027

If you work in equity research and see something wrong in the method, I'd like to hear it. The contact is on the cover.

19 How to run it yourself

Method 2 is public at github.com/Vidwaansinghania/Multi-Agent-Equity-Research under the MIT licence. There are two ways in.

Without any keys

The repository includes a complete worked run on an invented company, Acme Materials Corporation, so nothing in it is a view on a real stock. With Python 3.11 or newer you can install the requirements and run the three build scripts on that example to get the Word file, the 14-sheet Excel model and a six-page PDF. No API key, market-data account or Claude Code is needed.

On a real company

This needs Claude Code. Copy the example config file, add your name and email (the email goes into the SEC request header, without which EDGAR refuses every request), copy the Method 2 skill into Claude Code's skills folder, and type the trigger phrase with a ticker, for example "run a multi agent equity research report on MSFT". A market-data connector is optional; without one the market agent falls back to public price pages. Based on my three runs, expect roughly 19 to 22 million effective tokens, which is \$71 to \$88 at API prices, and two to three hours of active time spread over half a day or more.

What you'd need to change

The risk agent needs a holdings file to check calls against, so anyone else would need to supply their own; without one, the portfolio part of the check has nothing to work with. The rating bands, the DCF construction and the report template are written down in the repository's documentation.

20 What I learned

Some of this is about equity research, and some is about building with AI agents.

- Write the single-analyst version first and hold it to a real standard. The multi-agent version can only be as good as the rules it inherits.
- Reported numbers aren't the end of the analysis. Free cash flow, EPS and beta all turned out to need a second look.
- My most serious mistakes were in the parts I thought were routine: the ownership disclosures, a currency cost I never priced in, and a token count I got wrong by almost double.
- For agents, make the rating a fixed field, make the judge read the evidence, and keep the agents away from your own answer until they've committed to theirs.
- Check where the tokens go after the first run. The cost is in what agents carry on every step, not in what they look up.
- Log every call on the day it's made, including the ones you later withdraw.
- The first time the two methods disagreed, it came from the evidence stage building its own earnings estimate, not from the debate. Arguing is good for auditing a model; the estimate is what moves a rating.
- If I started again, I would set up the blind test from the first day, before I had any view of my own to leak into the agents.

References

1. Anthropic. (2026). Claude Opus 5 and Claude Sonnet 5 [large language models]. Model identifiers *claude-opus-5* and *claude-sonnet-5*, used June to September 2026. anthropic.com.
2. Anthropic. (2026). Claude Code [software]. claude.com/claude-code.
3. Xiao, Y., Sun, E., Luo, D., and Wang, W. (2024). TradingAgents: Multi-agents LLM financial trading framework. arXiv:2412.20138. Code: github.com/TauricResearch/TradingAgents.
4. U.S. Securities and Exchange Commission. EDGAR company filings and XBRL company facts. sec.gov/edgar.
5. Federal Reserve Bank of St. Louis. FRED series DGS10, 10-year Treasury constant maturity rate. fred.stlouisfed.org.
6. Alpha Vantage. Stock market data API. alphavantage.co.
7. CFA Institute. CFA Institute Research Challenge. cfainstitute.org.
8. Singhanian, V. (2026). Multi-Agent-Equity-Research, version as of September 13, 2026. github.com/Vidwaansinghanian/Multi-Agent-Equity-Research.

Company figures come from each company's own SEC filings and are cited in the individual reports, not repeated here.

A Build notes

Things that took real time to get right but don't affect the comparison between the two methods.

How the reports look

Report colours come from each company's own published brand colours, and every text and chart colour is checked for contrast before use. White text on Google Blue measures 3.56:1 and fails the 4.5:1 minimum, so the Alphabet header bar had to be near-black. White on Coca-Cola red passes at 4.67:1, and on Microsoft's blue at 5.38:1. The same work found that the colour-blindness simulation in my palette scripts was broken for one of the three types and returned the same colour for every input. The export step also strips my internal notes out of the finished files one sentence at a time, and I read the difference against the previous PDF every time, because the first pass on IREN deleted a whole paragraph I wanted to keep.

Smaller problems that cost time

- Connected tools show up with an ID and no name. I have seven market-data keys connected, each capped at 25 calls a day and shared between this project and a daily portfolio dashboard, so an agent told to use a named key had nothing to match on. On the first Microsoft pass the market agent made no calls rather than risk using another job's budget. The tools are also listed alphabetically by ID, which means nothing, and that's how my first guess at which key was which went wrong.
- SEC EDGAR rejects any request that doesn't include a name and email, so a standard web-fetch tool fails on every sec.gov page. Filings are downloaded with a script that sends that information.
- The serif font in my reports had never actually appeared in a PDF. Its font files use a format the Python PDF library can't read, so two months of reports had quietly fallen back to Helvetica. The build now prints which font it really used. This paper is built through a web browser, which reads the font correctly.

B Timeline and disclosure

Date, 2026	What happened
Mid-June	First Method 1 report written. ServiceNow BUY on June 20.
July 2	SEC EDGAR, the Alpha Vantage API and web extraction connected, along with report templates and the standards checklist.
August 8	Apple, IBM, IREN and Mogotes Metals reports. Every report switched to the company's own brand colours, with contrast checked.
August 9	Alphabet report. Ratings cut to three. Every report rebuilt and the ownership disclosures corrected.
August 10–16	Coca-Cola, Microsoft and Costco reports. Nine companies covered.

Date, 2026	What happened
August 17	Read TradingAgents and designed Method 2. Home Depot run on a nine-agent version, then restructured to ten agents with shared export scripts. Calls log started with the nine Method 1 calls. Microsoft run: HOLD, \$487.18.
August 18	Market-data keys mapped by ID. Method 2 written up as a public repository with a worked example on an invented company.
August 21–23	Coca-Cola Method 1 refresh, finished the evening of August 21, then a Method 2 run on the same closing price that API limits stretched to August 23: HOLD, \$82 and HOLD, \$77.45.
August 23	Coca-Cola token audit. Statement extraction, save-as-you-go and beta measurement added to the process and published.
September 7	Home Depot call withdrawn and archived; its row stays in the log.
September 13	Repository brought up to date with the live process. First version of this paper.
September 17	Token counts recounted from the session logs, which roughly halved the Coca-Cola total. Paper revised and restructured.
September 18	Paper expanded with the comparison design, the full list of debate claims, a list of mistakes and a timeline of next steps.
September 19–20	Method 2 run on Alphabet: SELL, \$268.81. Method 1's Alphabet node refreshed the next day off the same close: HOLD, \$340. The first disagreement between the two methods.
September 20	The Method 1 date for the ad-tech ruling corrected to September 2 and that node's exports rebuilt; the refreshed Alphabet call given its own row in the calls log.
September 21	Tools moved into a section of their own, with the model versions recorded, and the models and the harness added to the references.

Disclosure

At the time of the runs described here I owned shares of Microsoft and Coca-Cola, and of some of the other companies in section 6; each report's ownership section gives the position on its snapshot date. My current holdings are public on Blossom at blossomsocial.com/users/Vidwaan-Singhania_d7oVAMV5EWqGMPon. This paper describes a student research project built for learning. It is not investment advice, a recommendation to buy or sell any security, or a solicitation. The ratings and targets are dated, unscored and may already be out of date.

Code: github.com/Vidwaansinghania/Multi-Agent-Equity-Research, MIT licence, with a worked example on an invented company.